

SUNOJ RAGIRI

+91 93915 24412 nagasunoj777@gmail.com [linkedin.com/in/sunoj-ragiri97/](https://www.linkedin.com/in/sunoj-ragiri97/) github.com/sun-9545sunoj

Education

Indian Institute of Information Technology, Kottayam

Expected Graduation: 2027

Bachelor of Technology in Computer Science and Engineering

Kerala, India

– Relevant Coursework: Probability & Statistics, Linear Algebra, Data Structures & Algorithms, Optimization Techniques

Technical Experience / Open Source

QuantumLib/qsim (Google Quantum AI)

Jan 2026 – May 2026

GPU Systems Contributor [\[PR #1016\]](#)

Open Source

– Resolved GPU scaling bottlenecks, scaling state-vector simulator capacity beyond **34 qubits** via 64-bit memory indexing and 3D grid folding.

– Unified hardware-compliant grid logic in core C++/CUDA headers for cross-platform portability across NVIDIA and AMD (ROCm/HIP).

– Refactored compute kernels to eliminate integer overflows during massive address resolution; optimized kernel dispatch alignment.

Apache SeaTunnel (ASF)

Sep 2026 – Present

Distributed Systems Contributor [\[PR #10270\]](#)

Open Source

– Resolved distributed data-skew and checkpoint Exception timeout defects in the MySQL CDC connector during snapshot processing.

– Re-engineered chunk-splitting logic using dynamic sampling and distribution metrics to handle sparse primary keys,eliminate reader stalls.

– Implemented boundary-correctness and regression test suites using JUnit and Debezium mocks across multi-round ASF code reviews.

Projects

High-Frequency LOB Prediction & Execution Backtesting Framework – [Project Link](#)

Quantitative Market Microstructure & Deep Learning

Tech Stack: Python, PyTorch, PyTorch Lightning, Pandas, NumPy

• **Microstructure Feature Pipeline:** Engineered a high-throughput sliding-window engine in PyTorch Lightning processing 1.5M+ ticks across 5 equities, deriving 61 market microstructure signals (Order Flow Imbalance, bid-ask spread dynamics, depth ratios).

• **Predictive Deep Learning Models:** Implemented and evaluated DeepLOB, 1D-ResNet+Transformer, and spatial-temporal attention networks to forecast sub-second price moves from 10-level Limit Order Book (LOB) snapshots.

• **Market Simulation & Robustness:** Designed a domain-specific augmentation pipeline (Price Noise, Volume Scaling, Level Dropout) alongside focal loss to tackle heavy class imbalance and order book sparseness.

• **Covariate Shift Diagnostics:** Evaluated predictive alpha against out-of-sample regime shifts and distribution gaps across varying volatility environments, achieving an out-of-sample validation F1 score of ~0.76.

Financial Risk Analysis & Surveillance Pipeline – [Project Link](#)

Financial Risk Analytics & Production ML

Tech Stack: Python, Scikit-learn, XGBoost, LightGBM, Power BI

• **End-to-End Ingestion & Validation:** Built a modular Python pipeline automating batch ingestion, strict schema validation, IQR outlier handling, and automated feature transformation for credit and counterparty risk evaluation.

• **Risk Indexing & Metric Engineering:** Developed a weighted Financial Health Score and multi-variable exposure KPIs using Debt-to-Income, Asset Coverage, and leverage ratios.

• **Benchmarked Predictive Models:** Trained and evaluated Random Forest, XGBoost, and LightGBM models with SMOTE balancing on 30K+ transactional records, optimizing for precision-recall tradeoffs under asymmetric risk costs.

• **Surveillance & BI Reporting:** Integrated interactive dashboarding for portfolio risk tiering, exposure concentration, ROC evaluation, and feature attribution.

Technical Skills

Languages	Python, C++, SQL, R, Bash
Quantitative & Systems	Limit Order Book (LOB) Modeling, Market Microstructure, GPU Computing (CUDA/HIP), Backtesting
Machine Learning & AI	PyTorch, Scikit-learn, XGBoost, LightGBM, Transformers, Time-Series Modeling
Data & Processing	NumPy, Pandas, Feature Engineering, EDA, Data Validation Pipelines
Infrastructure & Tools	Docker, Kubernetes, FastAPI, Git, CI/CD, Linux, Prometheus, Grafana, GCP
BI & Visualization	Power BI, Matplotlib, Seaborn

Achievements & Certifications

- Selected as a **Google Student Ambassador 2026**
- Contributed merged pull requests to Google's QuantumLib (qsim) open-source repository
- WorldQuant University – Data Science Lab Certificate
- Google Cloud Skills Boost – Vertex AI & Generative AI